

2.3 CONTROLLER PILOT DATA LINK COMMUNICATION APPLICATION¹

2.3.1 INTRODUCTION

2.3.1.1 Overview

2.3.1.1.1 The CPDLC (CPDLC) application allows data link communication between controllers and pilots together with end-to-end protection of message integrity by an application level integrity check that also provides assurance of correct delivery.

2.3.1.1.2 The CPDLC application provides the capability to establish, manage, and terminate CPDLC dialogues between ATS ground and aircraft system peers. Once a dialogue is established, CPDLC provides for controller/pilot message exchange.

2.3.1.1.3 The CPDLC application also provides the capability to establish, manage, and terminate CPDLC dialogues between two ATC ground system peers for the purpose of ground/ground forwarding of a CPDLC Message.

Note 1.—Protection against mis-direction and corruption is not provided on ground forward messages.

Note 2.—Structure:

- a) 2.3.1: *INTRODUCTION* contains the structure of 2.3 and a summary of the functional capabilities of CPDLC.
- b) 2.3.2: *GENERAL REQUIREMENTS* contains the CPDLC version number, and error processing requirements.
- c) 2.3.3: *ABSTRACT SERVICE DEFINITION* contains the description of the abstract service provided by the CPDLC Application Service Element (CPDLC-ASE).
- d) 2.3.4: *FORMAL DEFINITION OF MESSAGES* contains the formal definition of messages exchanged by CPDLC ASEs using Abstract Syntax Notation Number One (ASN.1).
- e) 2.3.5: *PROTOCOL DEFINITION* describes the exchanges of messages allowed by the CPDLC protocol, as well as time constraints and CPDLC-ASE protocol descriptions and state tables.

¹PDR Applied:

PDR M5050003: Inclusion of PM-CPDLC Section 2.5

PDR M5080003: Positive PM-CPDLC-start-rsp with ERROR message element missing

PDR M5100001: PM-CPDLC - Incorrect notes on PM

PDR M5120001: Erroneous Air-ASE protocol state for CPDLC in SV-II

PDR M6010001: Security Parameter in PM-CPDLC

PDR M6050001: Single CPDLC

PDR M6060001: UM237 after extensibility marker

- f) 2.3.6: *COMMUNICATION REQUIREMENTS* contains the requirements that the CPDLC application imposes on the underlying communication system.
- g) 2.3.7: *CPDLC USER REQUIREMENTS* contains requirements imposed on the user of the CPDLC ASE service and message description tables.
- h) 2.3.8: *SUBSETTING RULES* defines the conformant subsets for the CPDLC-ASE.

Note 3.— Functional Descriptions

- a) The **Controller-Pilot Message Exchange Function** defines a method for a controller and pilot to exchange CPDLC messages via data link. This function provides CPDLC messages for the following :
 - 1) general information exchange;
 - 2) clearance
 - i) delivery,
 - ii) request, and
 - iii) response;
 - 3) altitude/identity surveillance;
 - 4) monitoring of current/planned position;
 - 5) advisories
 - i) request and
 - ii) delivery;
 - 6) system management functions; and
 - 7) emergency situations.
- b) The **Transfer of Data Authority Function** provides the capability for the current data authority to designate another ground system as the next data authority. A CPDLC dialogue can be opened with or by the next data authority at a time before becoming the current data authority. This capability is intended to prevent a loss of communication that would occur if the next data authority were prevented from actually setting up a dialogue with an aircraft until it became the current data authority. The designation of a next data authority is accomplished using a CPDLC message.

- c) The **Down Stream Clearance Function** provides the capability for an aircraft to contact an air traffic service unit which is not the current data authority for the purpose of receiving a down stream clearance. This information is exchanged using CPDLC message(s).
- d) The **Ground Forward Function** provides the capability for a ground system to forward information received in a CPDLC message to another ground system. The ground forwarding function can be used by the controlling data authority to forward an aircraft request to the next data authority, so that an aircraft does not need to issue the same request again. This function can also be used by a downstream data authority to pass a message to a current data authority for transmission by the current data authority to an aircraft. This information is exchanged using CPDLC message(s). It is a one-way forwarding of information with an indication of success, failure or non-support from the receiving ground system. The CPDLC Ground Forward function does not use the CPDLC Application Message Integrity Check.

Note 4.— See 2.3.7 for detailed CPDLC message intent/use descriptions.

2.3.2 GENERAL REQUIREMENTS

2.3.2.1 CPDLC ASE Version Number

2.3.2.1.1 The CPDLC-air-ASE and CPDLC-ground-ASE version numbers shall both be set to one.

2.3.2.2 Error Processing Requirements

2.3.2.2.1 In the event of information input by the CPDLC-user being incompatible with that able to be processed by the system, the CPDLC-user shall be notified.

2.3.2.2.2 In the event of a CPDLC-user invoking a CPDLC service primitive when the CPDLC-ASE is not in a state specified in 2.3.5, the CPDLC-user shall be notified.

2.3.3 THE ABSTRACT SERVICE

2.3.3.1 Service Description

2.3.3.1.1 An implementation of either the CPDLC ground based service or the CPDLC air based service shall exhibit external behavior consistent with having implemented a CPDLC-ground-ASE, or CPDLC-air-ASE respectively, with the following abstract service interface primitives, making them available to the CPDLC-ground-user or CPDLC-air-user respectively.

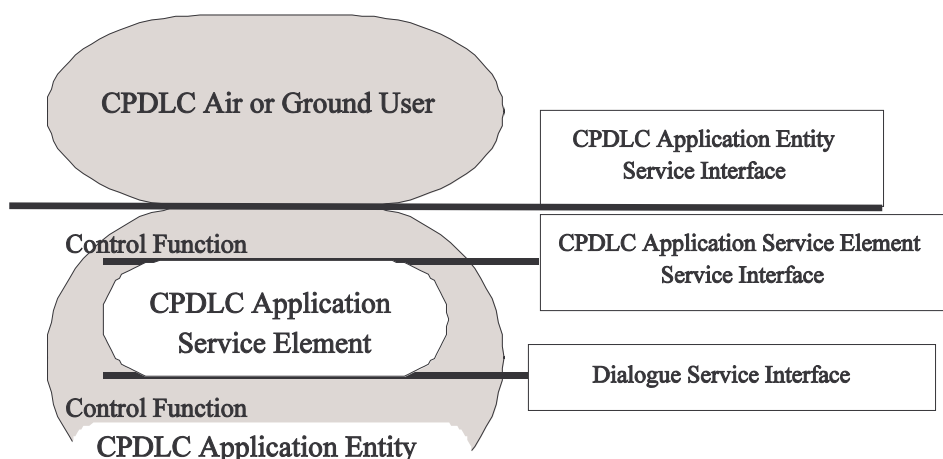
Note 1.— There is no requirement to implement the service in a CPDLC product; however, it is necessary to implement the ground based and air based system in such a way that it will be impossible to detect (from the peer system) whether or not the interface has been built.

Note 2.— This chapter defines the abstract service interface for the CPDLC service. The CPDLC-ASE abstract service is described in this chapter from the viewpoint of the CPDLC-air-user, the CPDLC-ground-user and the CPDLC-service-provider.

Note 3.— This chapter defines the static behavior (i.e., the format) of the CPDLC abstract service. Its dynamic behavior (i.e., how it is used) is described in 2.3.7.

Note 4.— Figure 2.3.3-1 shows the functional model of the CPDLC Application. The functional modules identified in this model are the following :

- a) the CPDLC-user,*
- b) the CPDLC Application Entity (CPDLC-AE) service,*
- c) the CPDLC-AE,*
- d) the CPDLC Control Function (CPDLC-CF),*
- e) the CPDLC Application Service Element (CPDLC-ASE) service,*
- f) the CPDLC-ASE, and*
- g) the Dialogue Service (DS).*



Note 5.—

The CPDLC-user represents the operational part of the CPDLC system. This user does not perform the communication functions but relies on a communication service provided to it via the CPDLC-AE through the CPDLC-AE service. The individual actions possible through the CPDLC-AE service are called service primitives.

Note 6.— The CPDLC-AE consists of several elements including the CPDLC-ASE and the CPDLC-CF.

Note 7.— The CPDLC-ASE is the element in the communication system which executes the CPDLC specific protocol. In other words, it takes care of the CPDLC specific service primitive sequencing, CPDLC Message creation, timer management, error and exception handling. The difference with the Standard Mode CPDLC is that the actual encoding and decoding of each CPDLC Message is handled by the CPDLC-users.

Note 8.— This CPDLC-CF is responsible for mapping service primitives received from one element (such as the CPDLC-ASE and the CPDLC-user) to service primitives of other abstract elements.

Note 9.— The CPDLC-ASE has two abstract boundaries with the CPDLC-CF: the CPDLC-ASE service and the dialogue service. The CPDLC-CF maps CPDLC-AE service primitives to other abstract elements in the CPDLC-AE and the underlying communication service, and vice versa.

2.3.3.2 The CPDLC-ASE Abstract Service

2.3.3.2.1 The CPDLC-ASE abstract service shall consist of a subset of the following services as allowed in 2.3.8:

- a) CPDLC-start service as defined in 2.3.3.3,
- b) DSC-start service as defined in 2.3.3.4,
- c) CPDLC-message service as defined in 2.3.3.5,
- d) CPDLC-end service as defined in 2.3.3.6,

- e) DSC-end service as defined in 2.3.3.7,
- f) CPDLC-forward service as defined in 2.3.3.8,
- g) CPDLC-user-abort service as defined in 2.3.3.9, and
- h) CPDLC-provider-abort service as defined in 2.3.3.10.

Note 1.— For a given primitive, the presence of each parameter is described by one of the following values in the parameter tables in 2.3.3.

- a) **blank** not present;
- b) **C** conditional upon some predicate explained in the text;
- c) **C(=)** conditional upon the value of the parameter to the left being present, and equal to that value;
- d) **M** mandatory;
- e) **M(=)** mandatory, and equal to the value of the parameter to the left;
- f) **U** user option.

Note 2.— The following abbreviations are used in this document:

- a) **Req** - request; data is input by CPDLC-user initiating the service to its respective ASE,
- b) **Ind** - indication; data is indicated by the receiving ASE to its respective CPDLC-user,
- c) **Rsp** - response; data is input by receiving CPDLC user to its respective ASE, and
- d) **Cnf** - confirmation; data is confirmed by the initiating ASE to its respective CPDLC-user.

Note 3.— An unconfirmed service allows a message to be transmitted in one direction without providing a corresponding response.

Note 4.— A confirmed service provides end-to-end confirmation that a message sent by one user was received by its peer user.

Note 5.— An abstract syntax is a syntactical description of a parameter which does not imply a specific implementation. Only when the CPDLC-ASE maps a parameter into an APDU field, or vice versa, the abstract syntax of the parameter is described by using ASN.1 of 2.3.4 for this field.

2.3.3.3 CPDLC-start Service

Note 1.— The CPDLC-start service is used by the CPDLC-air-user or CPDLC-ground-user to establish a CPDLC dialogue. It is a confirmed service.

Note 2.— Once a CPDLC dialogue is established it remains open until explicitly closed. (See CPDLC-end and CPDLC-abort services.)

2.3.3.3.1 The CPDLC-start service shall contain the primitives and parameters as presented in Table 2.3.3-1.

Table 2.3.3-1. CPDLC-start Service Parameters

Parameter Name	Req	Ind	Rsp	Cnf
Called Peer Identifier	M			
Calling Peer Identifier	M	M(=)		
Protected CPDLC Message	M	M(=)		
Response Message			M	M(=)
Result			M	M(=)
Class of Communication Service	U	M		
Security Required	U			

2.3.3.3.2 Called Peer Identifier

Note 1.— If the service is ground initiated, this parameter contains the addressed aircraft's 24-bit aircraft address.

Note 2.— If the service is air initiated, this parameter contains the addressed ground system's facility designation.

2.3.3.3.2.1 If the service is ground initiated, the *Called Peer Identifier* parameter value shall conform to the abstract syntax 24-bit aircraft address.

2.3.3.3.2.2 If the service is air initiated, the *Called Peer Identifier* parameter value shall conform to the abstract syntax four to eight-character facility designation.

2.3.3.3.3 Calling Peer Identifier

Note 1.— If the service is ground initiated, this parameter contains the sending ground system's facility designation.

Note 2.— If the service is air initiated, this parameter contains the sending aircraft's 24-bit aircraft address.

2.3.3.3.3.1 If the service is ground initiated, the *Calling Peer Identifier* parameter value shall conform to the abstract syntax four to eight-character facility designation.

2.3.3.3.3.2 If the service is air initiated, the *Calling Peer Identifier* parameter value shall conform to the abstract syntax 24-bit aircraft address.

2.3.3.3.4 Protected CPDLC Message

Note.— *The CPDLC-user uses this parameter to send a Protected CPDLC Message to its peer user.*

2.3.3.3.4.1 If the CPDLC-start request primitive is invoked by the CPDLC-ground-user, the *Protected CPDLC Message* parameter value shall conform to the ASN.1 abstract syntax ProtectedUplinkMessage.

2.3.3.3.4.2 If the CPDLC-start request primitive is invoked by the CPDLC-air-user, the *Protected CPDLC Message* parameter value shall conform to the ASN.1 abstract syntax ProtectedDownlinkMessage.

Note.— *A Protected CPDLC Message may contain no CPDLC Message.*

2.3.3.3.5 Response Message

Note.— *This parameter is used to either provide a reason for rejecting a CPDLC dialogue or to provide a means of verifying that a CPDLC dialogue establishment request has been received and accepted by the intended peer. In the latter case, the response includes either a LACK or no message element. This is to ensure a rapid response to a CPDLC Start. However, the CPDLC-ASE is not required to police this requirement. The air or ground CPDLC-user is instead required to ensure that the Response parameter includes either an allowed CPDLC Message or no CPDLC Message.*

2.3.3.3.5.1 If the CPDLC-start response primitive is invoked by the CPDLC-ground-user, the *Response Message* parameter value shall conform to the ASN.1 abstract syntax ProtectedUplinkMessage.

2.3.3.3.5.2 If the CPDLC-start response primitive is invoked by the CPDLC-air-user, the *Response Message* parameter value shall conform to the ASN.1 abstract syntax ProtectedDownlinkMessage.

Note.— *A Response may, and often does, contain no CPDLC Message.*

2.3.3.3.6 Result

Note.— *This parameter is used to indicate whether or not a requested CPDLC dialogue is accepted.*

2.3.3.3.6.1 This parameter shall have one of two abstract values: “accepted” or “rejected”.

2.3.3.3.7 Class of Communication Service

Note 1.— *This parameter contains the value of the required class of communication service. If not specified by the CPDLC-user, this indicates that there is no routing preference.*

Note 2.— *This parameter is used by the CPDLC-ground-user to determine if the Class of Communication value is acceptable for the establishment of a CPDLC dialogue.*

Note 3.— The parameter indicated to the peer CPDLC user is that provided by the CPDLC user if specified by the user, else it indicates that no routing preference was requested by the CPDLC dialogue initiator.

2.3.3.3.7.1 Where specified by the CPDLC-user, the *Class of Communication Service* parameter shall have one of the following abstract values: “A”, “B”, “C”, “D”, “E”, “F”, “G”, or “H”.

2.3.3.3.7.2 When this parameter is provided by the CPDLC-user, the same value shall be indicated to the peer CPDLC user, else the abstract value “ATSC - No Traffic Type Policy Preference” is indicated.

2.3.3.3.8 Security Required

Note 1.— The Security Required parameter contains the value of the required level of security, if specified by the CPDLC user.

Note 2.— If the received Security Required parameter is not as expected per the local security policy, the receiving CPDLC-ASE will abort.

2.3.3.3.8.1 Where specified by the CPDLC-user, the *Security Required* parameter shall have one of the following abstract values: “no security” or “secured exchange”.

Note.— Where not specified by the CPDLC-user, this indicates that there is no security required.

2.3.3.4 DSC-start Service

Note 1.— The DSC-start service is used by the CPDLC-air-user to establish a DSC dialogue for the purpose of providing down stream clearances. It is a confirmed service.

Note 2.— Once a DSC dialogue is established it remains open until explicitly closed. (See DSC-end and CPDLC-abort services.)

2.3.3.4.1 The DSC-start service shall contain the primitives and parameters as presented in Table 2.3.3-2.

Table 2.3.3-2. DSC-start Service Parameters

Parameter Name	Req	Ind	Rsp	Cnf
Facility Designation	M			
Aircraft Address	M	M(=)		
Protected CPDLC Message	M	M(=)		
Response Message			M	M(=)
Result			M	M(=)

Class of Communication Service	U	M		
Security Required	U			

2.3.3.4.2 Facility Designation

Note.— This parameter contains the addressed ground system's facility designation.

2.3.3.4.2.1 The *Facility Designation* parameter value shall conform to the abstract syntax four to eight-character facility designation.

2.3.3.4.3 Aircraft Address

2.3.3.4.3.1 The *Aircraft Address* parameter value shall conform to the abstract syntax 24-bit aircraft address.

Note.— This parameter contains the aircraft's 24 bit aircraft address.

2.3.3.4.4 Protected CPDLC Message

Note.— The CPDLC-air-user uses this parameter to send a Protected CPDLC Message to a CPDLC-ground-user.

2.3.3.4.4.1 The *Protected CPDLC Message* parameter value shall conform to the ASN.1 abstract syntax ProtectedDownlinkMessage.

Note.— A Protected CPDLC Message may contain no CPDLC Message.

2.3.3.4.5 Response Message

Note.— This parameter is used to either provide a reason for rejecting a DSC dialogue or to provide a means of verifying that a DSC dialogue establishment has been received and accepted by the intended peer. In the latter case, the response includes either a LACK or no message element. This is to ensure a rapid response to a DSC Start. However, the CPDLC-ground-ASE is not required to police this requirement. The CPDLC-ground-user is instead required to ensure that the Response parameter includes either a LACK or no CPDLC Message.

2.3.3.4.5.1 The *Response Message* parameter shall conform to the ASN.1 abstract syntax ProtectedUplinkMessage.

Note.— A Response may, and often does contain no CPDLC Message.

2.3.3.4.6 Result

Note.— This parameter is used to indicate whether or not a requested DSC dialogue is accepted.

2.3.3.4.6.1 The *Result* parameter value shall have one of two abstract values: "accepted" or "rejected".

2.3.3.4.7 Class of Communication Service

Note 1.— This parameter contains the value of the required class of communication service. If not specified by the CPDLC-air-user, this indicates that there is no routing preference.

Note 2.— This parameter is used by the CPDLC-ground-user to determine if the Class of Communication value is acceptable for the establishment of a DSC dialogue.

Note 3.— If provided by the user, the parameter indicated to the peer user is that provided by the user, else it indicates that no routing preference was requested by the CPDLC dialogue initiator.

2.3.3.4.7.1 Where specified by the CPDLC-air-user, the *Class of Communication Service* parameter shall have one of the following abstract values: “A”, “B”, “C”, “D”, “E”, “F”, “G”, or “H”.

2.3.3.4.7.2 When this parameter is provided by the CPDLC-user, the same value shall be indicated to the peer CPDLC-user, else the abstract value “ATSC - No Traffic Type Policy Preference” is indicated.

2.3.3.4.8 Security Required

Note 1.— The Security Required parameter contains the value of the required level of security, if specified by the CPDLC-air-user.

Note 2.— If the received Security Required parameter is not as expected per the local security policy, the receiving CPDLC-ground-ASE will abort.

2.3.3.4.8.1 Where specified by the CPDLC-air-user, the *Security Required* parameter shall have one of the following abstract values: “no security” or “secured exchange”.

Note.— Where not specified by the CPDLC-air-user, this indicates that there is no security required.

2.3.3.5 CPDLC-message Service

Note.— The CPDLC-message service can be used for pilot/controller message exchange, once a dialogue is established. It is an unconfirmed service.

2.3.3.5.1 The CPDLC-message service shall contain the primitives and parameters as presented in Table 2.3.3-3.

Table 2.3.3-3. CPDLC-message Service Parameters

Parameter Name	Req	Ind
Protected CPDLC Message	M	M(=)

2.3.3.5.2 Protected CPDLC Message

Note.— This parameter contains a Protected CPDLC Message.

2.3.3.5.2.1 If the CPDLC-message service is invoked by the CPDLC-ground-user, the *Protected CPDLC Message* parameter value shall conform to the ASN.1 abstract syntax ProtectedUplinkMessage.

2.3.3.5.2.2 If the CPDLC-message service is invoked by the CPDLC-air-user, the *Protected CPDLC Message* parameter value shall conform to the ASN.1 abstract syntax ProtectedDownlinkMessage.

Note.— This Protected CPDLC Message provided by the CPDLC-user always contains a CPDLC Message.

2.3.3.6 CPDLC-end Service

Note.— The CPDLC-end service is used by the CPDLC-ground-user to end a CPDLC dialogue with a CPDLC-air-user. It is a confirmed service.

2.3.3.6.1 The CPDLC-end service shall contain the primitives and parameters as presented in Table 2.3.3-4.

Table 2.3.3-4. CPDLC-end Service Parameters

Parameter Name	Req	Ind	Rsp	Cnf
Protected CPDLC Message	M	M(=)	M	M(=)
Result			M	M(=)

2.3.3.6.2 Protected CPDLC Message

Note.— This parameter contains a Protected CPDLC Message.

2.3.3.6.2.1 The *Protected CPDLC Message* parameter value shall conform to the ASN.1 abstract syntax ProtectedUplinkMessage, if provided by the CPDLC-ground-user.

2.3.3.6.2.2 The *Protected CPDLC Message* parameter value shall conform to the ASN.1 abstract syntax ProtectedDownlinkMessage, if provided by the CPDLC-air-user.

Note.— A Protected CPDLC Message may contain no CPDLC Message.

2.3.3.6.3 Result

Note.— This parameter is used to indicate whether or not a request to terminate a CPDLC dialogue is accepted.

2.3.3.6.3.1 The *Result* parameter shall have one of two abstract values: “accepted” or “rejected”.

2.3.3.7 DSC-end Service

Note.— The DSC-end service is used by the CPDLC-air-user to end a DSC dialogue with a CPDLC-ground-user. It is a confirmed service.

2.3.3.7.1 The DSC-end service shall contain the primitives and parameters as presented in Table 2.3.3-5.

Table 2.3.3-5. DSC-end Service Parameters

Parameter Name	Req	Ind	Rsp	Cnf
Protected CPDLC Message	M	M(=)	M	M(=)
Result			M	M(=)

2.3.3.7.2 Protected CPDLC Message

Note.— This parameter contains a Protected CPDLC Message.

2.3.3.7.2.1 The *Protected CPDLC Message* parameter value shall conform to the ASN.1 abstract syntax ProtectedUplinkMessage, if provided by the CPDLC-ground-user.

2.3.3.7.2.2 The *Protected CPDLC Message* parameter value shall conform to the ASN.1 abstract syntax ProtectedDownlinkMessage if provided by the CPDLC-air-user.

Note.— A Protected CPDLC Message may contain no CPDLC Message.

2.3.3.7.3 Result

Note.— This parameter is used to indicate whether or not a request to terminate a DSC dialogue is accepted.

2.3.3.7.3.1 The *Result* parameter shall have one of two abstract values: “accepted” or “rejected”.

2.3.3.8 CPDLC-forward Service

Note.— The CPDLC-forward service is used by a CPDLC-ground-user to send a CPDLC Message to another CPDLC-ground-user. Its primary use is for the forwarding of aircraft requests.

2.3.3.8.1 If the CPDLC-forward service is supported by the receiving ground system and the sending CPDLC-ground-ASE and receiving CPDLC-ground-ASE version numbers are equal, the CPDLC-forward service shall contain the primitives and parameters as presented in Table 2.3.3-6.

Table 2.3.3-6. CPDLC-forward Service Parameters (Service Supported, Versions Equal)

Parameter Name	Req	Ind	Cnf
Called Facility Designation	M		
Calling Facility Designation	M	M(=)	
CPDLC Message	M	M(=)	
Class of Communication Service	U		
Security Required	U		
Result			M

2.3.3.8.2 If the CPDLC-forward service is not supported by the receiving ground system, or if the CPDLC-forward service is supported by the receiving ground system but the sending CPDLC-ground-ASE and receiving CPDLC-ground-ASE version numbers are not equal, the CPDLC-forward service shall contain the primitives and parameters as presented in Table 2.3.3-7.

Table 2.3.3-7. CPDLC-forward Service Parameters (Service Not Supported or Versions Not Equal)

Parameter Name	Req	Cnf
Called Facility Designation	M	
Calling Facility Designation	M	
ASE Version Number		C
CPDLC Message	M	
Class of Communication Service	U	
Security Required	U	
Result		M

2.3.3.8.3 Called Facility Designation

Note.— This parameter contains the addressed ground system's facility designation.

2.3.3.8.3.1 The *Called Facility Designation* parameter value shall conform to the abstract syntax four to eight-character facility designation.

2.3.3.8.4 Calling Facility Designation

Note.— This parameter contains the sending ground system's facility designation.

2.3.3.8.4.1 The *Calling Facility Designation* parameter value shall conform to the abstract syntax four to eight-character facility designation.

2.3.3.8.5 ASE Version Number

Note.— This parameter contains the version number of the CPDLC-ASE.

2.3.3.8.5.1 When provided by the CPDLC-ground-ASE, the *ASE Version Number* parameter shall conform to the abstract integer value in the range 1-255.

2.3.3.8.5.2 Only if the sending CPDLC-ground-ASE version number is not equal to the receiving CPDLC-ground-ASE version number shall the receiving CPDLC-ground-ASE version number be confirmed to the sending CPDLC-ground-user.

Note.— If the sending CPDLC-ground-ASE version number is the same as the receiving CPDLC-ground-ASE version number, the Version Number parameter is not present in the indication given to the receiving CPDLC-ground-user, nor in the confirmation to the sending CPDLC-ground-user.

2.3.3.8.6 CPDLC Message

Note.— The sending CPDLC-ground-user uses this parameter to forward a CPDLC Message to another CPDLC-ground-user.

2.3.3.8.6.1 The *CPDLC Message* parameter value shall conform to the ASN.1 abstract syntax ATCForwardMessage.

2.3.3.8.7 Class of Communication Service

Note.— This parameter contains the value of the required class of communication service. If not specified by the CPDLC-ground-user, this indicates that there is no routing preference.

2.3.3.8.7.1 Where specified by the CPDLC-ground-user, the *Class of Communication Service* parameter shall have one of the following abstract values: “A”, “B”, “C”, “D”, “E”, “F”, “G”, or “H”.

2.3.3.8.8 Result

Note.— This parameter contains the result of the CPDLC-forward service. It will indicate success (service supported and matching versions), service unsupported, or version number incompatibility.

2.3.3.8.8.1 The *Result* parameter value shall conform to the ASN.1 abstract syntax ATCForwardResponse.

2.3.3.8.9 Security Required

Note 1.— The Security Required parameter contains the value of the required level of security, if specified by the CPDLC-ground-user.

Note 2.— If the received Security Required parameter is not as expected per the local security policy, the receiving CPDLC-ground-ASE will abort.

2.3.3.8.9.1 Where specified by the CPDLC-ground-user, the *Security Required* parameter shall have one of the following abstract values: “no security” or “secured exchange”.

Note.— Where not specified by the CPDLC-ground-user, this indicates that there is no security required.

2.3.3.9 CPDLC-user-abort Service

Note 1.— This service provides the capability for either the CPDLC-air-user or a CPDLC-ground-user to abort communication with its peer. It can be invoked at any time the CPDLC-user is aware that the CPDLC service is in operation. The CPDLC-user-abort service can be used for operational or technical reasons. It is an unconfirmed service. Messages in transit may be lost during this operation.

Note 2.— If the service is invoked prior to complete establishment of the dialogue, the CPDLC-user-abort indication may not be provided. A CPDLC-provider-abort indication may result instead.

2.3.3.9.1 The CPDLC-user-abort service shall contain the primitives and parameters as presented in Table 2.3.3-8.

Table 2.3.3-8. CPDLC-user-abort Service Parameters

Parameter Name	Req	Ind
Reason	U	M

2.3.3.9.2 Reason

Note 1.— This parameter is used to indicate a reason for aborting the CPDLC or DSC dialogue.

Note 2.— If provided by the CPDLC-user, the parameter indicated to the peer CPDLC-user is that provided by the CPDLC-user, else it is what the ASE supplies.

2.3.3.9.2.1 The *Reason* parameter value shall conform to the ASN.1 abstract syntax CPDLCUserAbortReason.

2.3.3.9.2.2 When this parameter is provided by the CPDLC-user, the same value shall be indicated to the peer CPDLC-user.

2.3.3.10 CPDLC-provider-abort Service

Note.— This service provides the capability for the CPDLC-service provider to inform its active users that it can no longer provide the CPDLC service. Messages in transit may be lost during this operation.

2.3.3.10.1 The CPDLC-provider-abort service shall contain the primitives and parameters as presented in Table 2.3.3-9.

Table 2.3.3-9. CPDLC-provider-abort Service Parameters

Parameter Name	Ind
Reason	M

2.3.3.10.2 Reason

Note.— This parameter identifies the reason for the abort.

2.3.3.10.2.1 The *Reason* parameter shall conform to the ASN.1 abstract syntax CPDLCProviderAbortReason.